

FPGA-Based Traffic Sign Detection Using Canny Edge Detection and Hough Transform

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Abstract— The Traffic Sign detection system is considered to be an onboard safety feature after the driver whose role is to detect and interpret road signs like stop, yield, and speed limit. An important part of advanced driver assistance systems (ADASs) stands for enhancing the awareness of the driver in terms of reducing the chances of human errors and ensuring the safety of everyone on the road. This paper gives a description of the design and implementation of the real-time traffic sign detection system using Field Programmable Gate Arrays (FPGAs). The proposed system captures live video input from the camera, then processes each video frame by applying red color segmentation on the YCbCr color space followed by Canny edge detection, which helps to trim down the sign boundaries and lastly uses the Hough transform to accurately detect geometric shapes corresponding to traffic signs. Using the benefits of parallel processing and reconfigurability of FPGAs, the system is capable of achieving high-speed and efficient detection of signs in real-time vehicular environments. The hardware implementation carried out over an Altera Cyclone IV-E FPGA (DE2-115 board) determined high accuracy of detection and low latency. The experimental results have proven that the system is able to detect traffic signs reliably in varying light conditions, minimize false positives, and address dynamic background interferences. This work further actively supports the move toward developing robust real-time embedded vision systems for next-generation autonomous and driver-assist vehicles.

Keywords— *FPGA, Traffic Sign Detection, Canny Edge Detection, Hough Transform, Real-time Video Processing, DE2-115.*

I. INTRODUCTION

In recent years, with respect to mainly vision-based advanced driver assistance systems, the sizes have increased primarily in the real-time recognition and interpretation of road signs. The detection of traffic signs in an accurate and very early way is very critical for any form of regulatory, warning, and informational signals to be communicated to the driver and an autonomous system. Hence, it contributes to a smaller number of accidents and improves overall road safety. Conventional software-based implementations of traffic sign detection usually consist of running hardware to detect signs on general-purpose processors (GPPs) or graphics processing units. These systems have problems concerning the speed of processing, power throughput. He and Yuan (2008) [7], on the other hand, designed improvements toward efficiency on FPGAs by optimizing kernel

consumption, and latency, holding them back from being incorporated into embedded automotive applications, which require low power and low latency.

Field-Programmable Gate Arrays (FPGAs) have now stood out as a very attractive alternative, providing a flexible hardware platform on the basis of a fully parallel processing system with deterministic latency and reconfigurability. FPGAs can also be used for rapid prototyping combined with iterative design unlike ASICs, which are specific declarations to fixed functions and are just designed for use, with no possible modification once manufactured. Such flexibility is very useful in dynamic environments, for example, in traffic monitoring systems, which require continuous updates in the detection algorithms to accommodate variations in road condition and changes in the background environment.

More recently, some successful works have been done on traffic sign detection and recognition using FPGA-based implementations. For example, Schwegelshohn et al. (2015) proposed an FPGA-based real-time system for automotive camera applications which exhibited very robust performance under difficult conditions. Shi et al. (2017) developed a power-hungry hardware accelerator through the use of Viola-Jones classifiers, which increased efficiency in detection throughput. Kim et al. (2022) introduced a real-time traffic sign classification using a very cheap fully integer-based convolutional neural network on FPGA recently. In addition, Lin et al., [4] benchmarked several deep learning frameworks for their FPGA deployment by discussing the trade-offs between latency and power efficiency in designs, while Zhou et al. [5], indicated a system-on-chip design towards an SoC approach for real-time traffic-sign recognition on modular, scalable architectures.

Edge detection is considered a very important image processing step when it comes to recognizing traffic signs. Canny edge detection, as formulated by Canny, is still widely used because of its superior performance in delineating an object boundary. Xu et al. (2014) [6] present a distributed version of Canny's algorithm, making it fit for hardware constraints with maximum

computations and memory management techniques. Tan and He (2020) [8] have made last strides advancing the real-time

implementation of Canny edge detection on FPGA for video streams. This serves to recognize its present-day embedded applications.

In addition to performing edge detection, the successful recognition of traffic signs often involves a look into the shape and geometry, for which the Hough Transform has been considered standard. Chen et al. (2012) [9] present an ultra-low resource-consuming architecture to carry out Hough transforms on FPGA, wherein they showcased the detection of circular features with the minimal latency and resource occupancy. However, despite these good advancements, many implemented systems today cannot handle factors such as dynamic lighting, background image clutter, and color variations that undermine detection reliability.

This paper proposes systems that will combine many custom IP cores, each concentrating on a certain task within the detection pipeline. The design consists of a custom red color segmentation module in the YCbCr color space to segment potential traffic sign areas, followed by the Canny edge detection core optimized for hardware that refines these areas by accurately accessing edge information. This is then followed by the custom Hough Transform IP core that recognizes the geometric shapes of the detected traffic signs. The whole processing chain runs on the Altera DE2-115 development board, ensuring the low-latency, real-time digital implementations are performed even under difficult environmental conditions using DSP blocks and logic elements provided by Cyclone IV-E FPGA.

The remainder of the paper is organized as follows: Section II discusses the detection algorithm, Section III outlines the methodology and design approach, Section IV presents the hardware implementation, Section V includes experimental results and analysis, and Section VI concludes the paper with future directions and enhancements.

II. SIGN DETECTION ALGORITHM

The suggested traffic sign detection unit implements a multi-stage processing pipeline onto an FPGA. In short, this processing pipeline consists of three main stages: red color segmentation, edge detection, and shape recognition. Each stage is realized as a custom IP core integrated onto the FPGA, allowing real-time processing with low latency.

The input video frames are transformed from RGB to the YCbCr color space, which proves to be very good for segmentation since the luminance (Y) and chrominance (Cb and Cr) are separated, making the algorithm more robust against lighting conditions. At this stage, the custom Red Color Detection IP Core uses preset threshold values to apply to the Cb and Cr channels so as to separate out regions that have a chance of containing red traffic signs. Pixels that have chrominance values confined within the specified range are the only ones marked as possible candidates,

thus, giving rise to a binary mask that highlights areas of interest.

After segmentation, the Canny Edge Detection IP Core is run on the binary mask. The Canny algorithm smooths the image with a Gaussian filter to reduce the noise, followed by computing the gradient intensity, aiding to distinguish the boundaries of the regions. A key step during this stage is the calculation of gradient magnitude, given by a simplified equation:

$$M(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2} \quad (1)$$

where G_x and G_y denote the gradients in the horizontal and vertical directions. The present equation conveys the essence of the algorithm adopted in the Canny edge detection module in highlighting the characteristic edges of the traffic signs.

In the last step, geometric features are extracted using the Hough Transform IP Core, which directs the edge pixels into parameter space to detect circular or triangular shapes typical of traffic signs. For the circular shape, the standard equation for a circle is applied:

$$(x - a)^2 + (y - b)^2 = r^2 \quad (2)$$

where (a,b) denote the center of the circle while r represents its radius. Increasing votes based on various radii and centers in the parameter space will then keep providing the possible geometric pattern for further defining the probable traffic sign candidates.

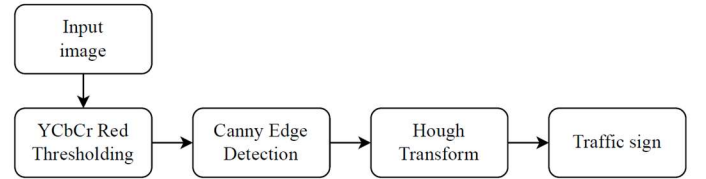


Fig. 1 Sign Detection Algorithm

III. METHODOLOGY

The proposed traffic sign detection system is built using the DE2-115 FPGA development board and Quartus Prime software. Its architecture is inherently suitable for real-time video processing using a handycam input. This work mainly aims to achieve hardware-accelerated image analysis by disintegrating the whole video pipeline into modular, reusable custom IP blocks for color segmentation, edge detection, and shape recognition. The complete flow of the system is described in Fig 2, which sketches the step-by-step signal processing pipeline from video acquisition to traffic sign identification and VGA display output.

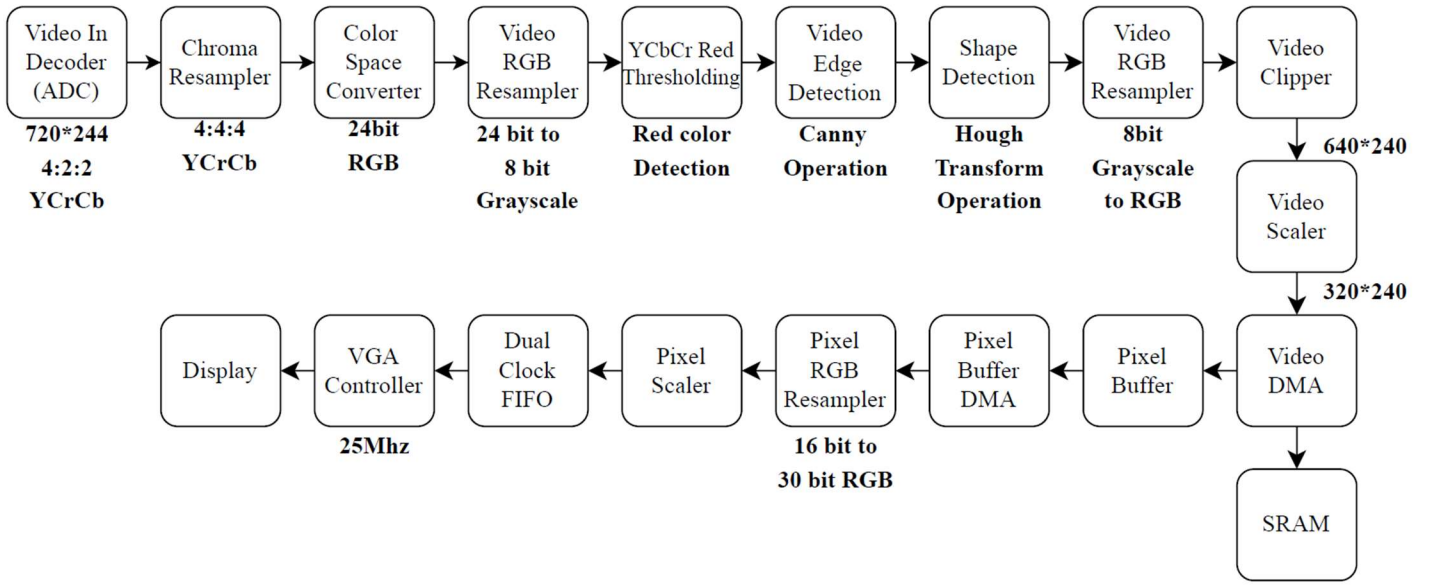


Fig 2: Complete video processing flow of Collision avoidance system

The raw video feed is taken using a handycam camera and digitized by an external video decoder which streams YCrCb 4:2:2 formatted frames to the FPGA board. These are first up sampled to YCrCb 4:4:4, then converted to 24-bit RGB format for compatibility with the processing modules.

This conversion from YCrCb to RGB is done using the standard ITU-R BT.601 conversion equations:

$$\begin{aligned}
 R &= Y + 1.402 (Cr - 128) \\
 G &= Y - 0.71414(Cr - 128) - 0.34414(Cb - 128) \\
 B &= Y + 1.772 (Cb - 128)
 \end{aligned} \quad (3)$$

Every frame in RGB is passed on to the Red Color Detection IP, which separates its regions of interest by converting the RGB pixel stream to the HSV color space and applying the thresholds for hue, saturation, and value that describe the red spectrum. The calculation of HSV values is done internally through a logic approximating:

$$\begin{aligned}
 V &= \max(R, G, B), \\
 S &= \frac{V - \min(R, G, B)}{V}, \\
 H &= \text{function of } R, G, B
 \end{aligned} \quad (4)$$

Once the red areas have been successfully segmented, the converted binary maps are passed to the Canny Edge Detection IP, which is responsible for marking the edges corresponding to potential traffic signs. The Canny algorithm is implemented completely in hardware and consists of four basic steps: Gaussian smoothing, gradient computation, non-maximum suppression, and hysteresis thresholding. The Gaussian smoothing operation is performed using a 2D Gaussian filter to suppress high-frequency noise:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (5)$$

The resulting image is then convoluted both in directions and y, with the Sobel operators to compute the gradient magnitude and orientation:

$$G = \sqrt{G_x^2 + G_y^2} \quad (6)$$

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right) \quad (7)$$

Where G_y and G_x were determined through the use of 3×3 convolution kernels. Consequently, sharp edge detection is ensured along the traffic sign contours. After edge detection, a third module, the Hough Transform IP, is adapted in the detection of geometric shapes like triangles, circles, and rectangles that are generally linked with regulatory and warning traffic signs. The Hough Transform is designed to accumulate votes in parameter space on circular form, being the voting scheme in consideration from (2).

Where (x,y) is the edge pixel from a binary edge map, while (a,b) are the center coordinates considered for the particular radius r. Detection ends by determining local maxima in this 3D accumulator space. The full architecture is created in Verilog HDL and synthesized onto the DE2-115 board using Quartus Prime. Video output is routed to VGA display through a VGA controller IP and real-time sign detection status. The design also benefits from dual-port SRAM buffers and FIFO modules for real-time frame memory and synchronized processing across pixel and system clocks.

This multi-stage hardware-based approach would enable a high-speed low latency detection of traffic signs suitable for embedded automotive applications. Because an FPGA performs processing in parallel, such a system does not endure the usual performance barriers suffered by systems based on traditional

software, and is therefore capable of continuous, real time operation at very high input frame rates.

IV. HARDWARE UTILIZATION

The Cyclone IV-E FPGA board supports the attempt and implementation of the proposed algorithm written in Verilog HDL. This hardware architecture is an ideal implementation of a collision avoidance algorithm with real-time video input since it offers massive parallel processing ability along with considerable I/O support. Plenty of DSP elements and embedded memory are provided on the FPGA board to carry out multi-frame iterative video processing. The integrated VGA output can render video for an essential analysis of video input in real-time.

Hardware components used

- Altera's Cyclone IV-E DE2-115 FPGA
- Sony HDR XR520V AVCHD Camcorder
- Monitor with VGA support

V. RESULTS

The development and testing of the FPGA-based traffic sign detection system was successfully implemented using the DE2-115 board of an FPGA. The subtutling on this board was the live capture of real-time road scenes via Handycam camera and feeding them into the FPGA through video-in interface. The capture is then processed frame by frame through a custom designed sequence of IP cores.

Fig.3(a) Original traffic scene captured via Handycam: In this test case. This raw input video will first pass through the red color detection IP core. As can be seen in Fig.3(b), the output effectively highlights only the red regions in the frame. This ensures the separation of traffic signs, where a majority of them are predominantly red.

Thus, the result goes further to the Canny edge detection IP core. The outcome of the operation is shown in Fig.3(c), where it shows the edges of the red detected area, making its circular shape clear for the sign. This would make clear the required definition of boundaries for the later detection.

Lastly, a Hough Transform IP core is used to detect certain geometrical shapes, in this case, the circles. Fig.3(d): Successful detection of the circular boundary of the speed limit sign which showcases that the system is capable of identifying the appropriate traffic signs accurately.

Fig. 4 presents the complete hardware setup used for the real-time traffic sign detection system, including the Handycam camera input and output display on the monitor, along with experimental results implemented on the DE2-115 development board featuring Altera's Cyclone IV-E FPGA.

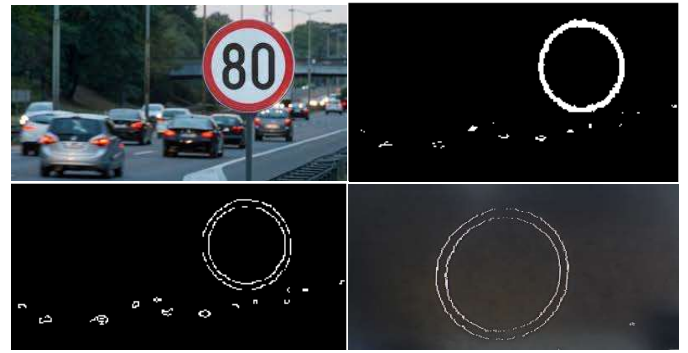


Fig 3. a. Original image, b. Red color detection, c. Canny edge detection, d . Hough Transform



Fig 4. Hardware setup and Video output

VI. CONCLUSION

This work describes the working of a real-time traffic sign detection system built on the DE2-115 FPGA board, operating in a hardware-accelerated video processing pipeline. The system features multiple custom IP cores developed in Verilog HDL for essential operations such as red-color segmentation, Canny-edge detection, and Hough Transform-based shape detection. The modules worked in the sequence of isolating areas occupying the red color, detecting strong edges, and recognizing circular patterns matching the expected prototypes of common traffic signs. A Handycam camera was the video source to assess the conditions practically, while the outputs were shown on VGA in real time, thus validating the accuracy and the quick response of the design. The results reveal the ability of the system to detect traffic signs with high reliability, low latency, and without any external processor or software involvement, therefore forming a good candidate for emergency-wise scenarios. The use of FPGA establishes parallel execution with deterministic performance, which makes it a good candidate for embedded and automotive applications demanding a real-time decision-making capability. Such a success story shows that FPGA technology could be effectively harnessed in intelligent transportation systems and could be taken further to establish recognition of multiple classes of signs, integration of machine learning-based classification, or support of vehicle to infrastructure systems in the coming development tracks.

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